An Ultraviolet Overlap Benchmark for a Nonlinear SU(2) Lattice Block

Heat-kernel tails, Haar Jacobians, and the first dependent-measure obstruction

Mathematical research dossier

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Abstract

The preceding nonlinear block certificate controlled a very small local chart, but at heat time $t = 10^{-2}$ almost every standalone heat-kernel plaquette lay outside it. We determine whether that mismatch is structural or merely a scale mismatch. First, a cancellation-free Poisson-image estimate bounds the standalone SU(2) curvature tail uniformly at weak coupling. Second, Euler's product for $\sin r/r$ gives explicit Hessian and third-derivative bounds for the link Haar Jacobian. Third, an independent connected-set theorem converts a one-plaquette tail into a weighted block activity benchmark.

At

$$t = 3 \cdot 10^{-6}, \qquad r = 0.0041, \qquad \rho = 2r = 0.0082, \qquad r_F = 10^{-5},$$

all local-action inequalities remain strict after the Haar contribution. The standalone tail is at most $5.51231 \cdot 10^{-5}$; after a 216-plaquette union bound, the weighted independent-block branching factor is at most 0.842163 and the rooted weighted activity sum is at most 0.083370. Thus local nonlinear control and small ultraviolet defect activities are not intrinsically incompatible.

This is a restricted coexistence certificate, not full-measure Yang–Mills control. Plaquettes in the four-dimensional link measure share links, obey nonabelian constraints, and carry boundary and topological information. The standalone heat-kernel law is not proved here to be their marginal, and block independence is false. The first unproved implication is therefore a volume-uniform dependent activity theorem for the complete gauge-invariant link measure.

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1 Question, conventions, and result

We use the unit-three-sphere model of SU(2) and heat generator $\frac{1}{2}\Delta$. A class angle is $\theta \in [0, \pi]$, and the leading small-time action is $\theta^2/(2t)$. The side-two four-dimensional cubical block has 216 links and 216 plaquettes. Stage 08 established the nonlinear plaquette-log and local Coulomb-chart constants on this block; its exact quaternion calculus is the finite geometric input used below.

The distinction between three measures is essential.

1. The *standalone heat-kernel law* has class density $p_t(\theta)(2/\pi)\sin^2\theta\,\mathrm{d}\theta$.
2. The *independent-block benchmark* assigns independent bad events with a declared block-probability upper bound.
3. The *four-dimensional link measure* integrates link variables against a product of plaquette weights and link Haar measures. Its plaquette variables are constrained and dependent.

Only the first two enter the probability theorem proved here. The local action inequalities, by contrast, concern the actual finite block action as a function of its link coordinates.

Theorem 1.1 (Restricted ultraviolet overlap witness)**.** *Let*

$$t = 3 \cdot 10^{-6}, \quad r = 0.0041, \quad \rho = 0.0082, \quad r_F = 10^{-5}, \quad \Delta = 8, \quad \alpha = 0.1.$$

Then the following statements hold.

1. *On the Stage 08 logarithmic Coulomb chart, the nonlinear physical singular-value lower bound is* 1.332871*.*
2. *After adding the exponential-link Haar density, the scaled Hessian remainder is at most* $0.968684 < 1$*, the centered fiber transport coefficient is at most* $0.003941 < 1$*, and the generated reference-base margin is at least* $0.060732 > 0$*.*
3. *For a standalone heat-kernel plaquette,*

$$\Pr_t\{\theta \geq \rho\} \leq 5.512301 \cdot 10^{-5}.$$

4. *Let* p_B *be the union-bound proxy for 216 such plaquette failures. Then* $p_B \leq 0.0119066$*. For independent bad blocks on a graph of maximum degree eight,*

$$e^{\alpha}\Delta^2 p_B \leq 0.842163 < 1$$

and

$$\sum_{X \ni o} e^{\alpha|X|} \Pr\{X \text{ is all bad}\} \leq 0.083370 < 1.$$

Every displayed pass inequality is strict; hence their simultaneous validity persists on a nonempty open neighborhood of (t, r, r_F)*.*

The proof is assembled in Sections 2–5.

Remark 1.2 (Logical status)**.** Part 4 is an independent-block theorem. It is not applied to the dependent four-dimensional link measure. Theorem 1.1 proves numerical and analytic coexistence of the two required regimes, not basin entry for the complete theory.

2 The link Haar Jacobian

For $A \in \mathbb{R}^3$ with $a = \|A\| < \pi$, exponential coordinates pull normalized Haar measure back, up to a constant independent of A, with radial Jacobian

$$J(A) = \left(\frac{\sin a}{a}\right)^2.$$

Its contribution to the coordinate action is

$$h(A) = -\log J(A) = -2\log\frac{\sin a}{a}.$$

Lemma 2.1 (Local Haar derivative bounds)**.** *For $0 \le a \le r < \pi$, put $x = r/\pi$ and $q = x^2$. Then*

$$\left\|D^2 h(A)\right\| \le H_2(r) := \frac{2}{3}\frac{1+q}{(1-q)^2}, \tag{1}$$

$$\left\|D^3 h(A)\right\| \le H_3(r) := \frac{2x}{3\pi}\left(\frac{4(1+q)}{(1-q)^3} + \frac{2}{(1-q)^2}\right). \tag{2}$$

At $r = 0.0041$ these bounds are $H_2 \le 0.666671$ and $H_3 \le 0.001662$.

Proof. Euler's product and absolute convergence on compact subsets of $(-\pi, \pi)$ give

$$h(a) = 2\sum_{n\ge1}\frac{\zeta(2n)}{n\pi^{2n}}a^{2n}.$$

For $p = 2n$, the Hessian operator norm of $\|A\|^p$ is at most $p(p-1)a^{p-2}$. Direct differentiation gives the third tensor as one radial cubic term and three mixed terms; the triangle inequality bounds its operator norm by $p(p-1)(p-2)a^{p-3}$. Since $\zeta(2n) \le \zeta(2) = \pi^2/6$, termwise differentiation yields

$$\left\|D^2 h\right\| \le \frac{2}{3}\sum_{n\ge1}(2n-1)q^{n-1} = \frac{2}{3}\frac{1+q}{(1-q)^2}.$$

Likewise, after writing $m = n - 1$,

$$\left\|D^3 h\right\| \le \frac{2}{3\pi x}\sum_{m\ge1}(2m+1)(2m)q^m,$$

and the standard sums for $\sum mq^m$ and $\sum m^2q^m$ give (2). □

The scaled action used in the local norm is tS. Since the link Haar potential is a sum of one-link functions, its global Hessian and third-tensor bounds are $tH_2(r)$ and $tH_3(r)$; no plaquette-count factor is introduced. At the witness these contributions are $2.00002 \cdot 10^{-6}$ and less than $5 \cdot 10^{-9}$, respectively.

3 A cancellation-free standalone tail

The character expansion is exact but becomes a numerically ill-conditioned alternating representation when t is extremely small. The Poisson-image representation exposes the ultraviolet concentration directly. Up to a positive factor independent of θ, the heat-kernel shape is

$$p_t(\theta) \propto \frac{1}{\sin\theta}\sum_{k\in\mathbb{Z}}(\theta + 2\pi k)\exp\left[-\frac{(\theta+2\pi k)^2}{2t}\right]. \tag{3}$$

Together with radial Haar density, the unnormalized radial mass is therefore $\sin\theta$ times the image sum.

Proposition 3.1 (Image tail upper bound)**.** *Let $0 < \rho < \pi$, $t > 0$, and $L = \min\{\pi/2, 3\sqrt{t}\}$. Define*

$$D_0(t,L) = \frac{\sin L}{L}\left[\sqrt{\frac{\pi}{2}}t^{3/2}\operatorname{erf}\left(\frac{L}{\sqrt{2t}}\right) - tLe^{-L^2/(2t)}\right],$$
$$N_0(t,\rho) = t\rho e^{-\rho^2/(2t)} + t\sqrt{\frac{\pi t}{2}}\operatorname{erfc}\left(\frac{\rho}{\sqrt{2t}}\right),$$
$$E(t) = t\frac{e^{-\pi^2/(2t)} + e^{-2\pi^2/t}}{1 - e^{-4\pi^2/t}}.$$

If $D_0(t,L) > E(t)$, then the standalone heat-kernel law satisfies

$$\Pr_t\{\theta \geq \rho\} \leq \min\left\{1, \frac{N_0(t,\rho) + E(t)}{D_0(t,L) - E(t)}\right\}. \tag{4}$$

Proof. For the zero image, $\sin\theta \leq \theta$ and extension of the upper integration limit to infinity give

$$\int_\rho^\pi \theta\sin\theta e^{-\theta^2/(2t)}\,\mathrm{d}\theta \leq \int_\rho^\infty \theta^2 e^{-\theta^2/(2t)}\,\mathrm{d}\theta = N_0(t,\rho).$$

On $[0,L]$, concavity of sine gives $\sin\theta \geq (\sin L/L)\theta$, so the retained zero-image normalization is at least $D_0(t,L)$.

It remains to bound every nonzero image in absolute value. For the negative jth image, $j \geq 1$, substitution gives an integral bounded by $te^{-(2j-1)^2\pi^2/(2t)}$; for the positive image the bound is $te^{-2j^2\pi^2/t}$. Consecutive terms in both sequences have ratio at most $e^{-4\pi^2/t}$. Summing the geometric majorant gives $E(t)$. Adding E to the tail numerator and subtracting it from the normalization denominator proves (4). □

At the witness, $\rho/\sqrt{t} = 4.734272$ and the nonzero-image term is smaller than $10^{-714000}$. Formula (4) gives $5.512301 \cdot 10^{-5}$. An independent character calculation truncated at representation index 6000 encloses the actual tail between $5.3507587609 \cdot 10^{-5}$ and $5.3507587704 \cdot 10^{-5}$, providing a cross-representation check without entering the proof of the upper bound.

4 The restricted weighted-polymer theorem

Theorem 4.1 (Independent rooted connected sets)**.** *Let a graph have maximum degree Δ. Suppose vertices are bad independently with probability at most p_B. For $\alpha \geq 0$, if $q_\alpha = e^\alpha\Delta^2 p_B < 1$, then*

$$\sum_{\substack{X \ni o \\ X \text{ connected}}} e^{\alpha|X|}\Pr\{X \text{ is all bad}\} \leq \frac{e^\alpha p_B}{1 - q_\alpha}. \tag{5}$$

Proof. A depth-first traversal encodes a rooted connected n-set by a walk of length at most $2(n-1)$, so the number of such sets is at most $\Delta^{2(n-1)}$. Independence bounds the all-bad probability by p_B^n. Summing the resulting geometric series proves (5). □

For a block containing 216 plaquettes, the union bound gives $p_B \leq 216p$. No within-block independence is needed for this union bound. Independence between bad blocks is, however, indispensable for Theorem 4.1; it is the assumption that fails in the target four-dimensional gauge measure.

5 Composition and numerical margins

Let η_P and τ_P be the Stage 08 scaled plaquette-action remainder and third-tensor bound. Lemma 2.1 gives the augmented quantities

$$\eta = \eta_P + tH_2(r), \qquad \tau = \tau_P + tH_3(r).$$

In the same Hodge-whitened retained/fiber coordinates, with reference curvatures $\lambda_F = 2$, $\lambda_Y = 1$, the five-component bounds become

$$\varepsilon_F = \eta, \quad \varepsilon_{FY} = \frac{\eta}{\sqrt{2}}, \quad \varepsilon_Y = \frac{\eta}{2}, \quad \tau_F = \tau, \quad \tau_{FY} = \frac{\tau}{\sqrt{2}}.$$

Writing $m_F = 2 - \eta$, the response and generated-action formulas are

$$K_F = \frac{r_F}{m_F}\left(\frac{\tau}{\sqrt{2}} + \tau\frac{\eta/\sqrt{2}}{m_F}\right),$$

$$\eta_{\mathrm{gen}} = \frac{\eta}{2} + \frac{(\eta/\sqrt{2})^2}{m_F}, \qquad m_Y = 1 - \eta_{\mathrm{gen}}.$$

Substitution gives the values in Table 1.

Table 1: Certified witness values.

Quantity	Bound	Test
η_P	0.968681008	local plaquette input
$tH_2(r)$	0.000002001	Haar contribution
η	0.968683008	< 1
m_F	1.031316992	> 0
K_F	0.003940979	< 1
m_Y	0.060732040	> 0
p	0.0000551231	standalone tail
p_B	0.011906570	$216p$
$e^{\alpha}\Delta^2 p_B$	0.842162811	< 1
rooted weighted sum	0.083369414	< 1

This proves Theorem 1.1. The strict-continuity conclusion follows because the denominators in the local inverse, fiber response, image normalization, and polymer sum remain positive at the witness.

For comparison, at the Stage 08 point $(t, r) = (10^{-2}, 0.003)$, the exact standalone probability outside $\rho = 0.006$ exceeds 0.9999. The local action remains controlled, but the independent activity proxy fails completely. The successful ultraviolet witness shows that this failure was not a universal chart-versus-probability incompatibility.

6 What the computation does and does not certify

The executable checker evaluates the exact displayed formulas, performs a declared 25×25 parameter-grid calibration, and runs positive, boundary, monotonicity, cross-representation, and deliberately failing tests. Of 625 grid points in

$$t \in [10^{-6}, 5 \cdot 10^{-6}], \qquad r \in [0.0036, 0.0042],$$

389 pass the restricted inequalities with $r_F = 10^{-5}$. This demonstrates that the witness is not an isolated floating-point accident. The grid search does not claim global optimality, and no proof depends on the selected grid.

The mechanization boundary is equally important. The checker does not formalize the analytic Poisson summation identity, the Stage 08 nonlinear inverse theorem, or any infinite-volume cluster expansion. It checks their closed finite consequences and fails closed on invalid radii, nonpositive normalizations, exhausted coercivity, and divergent polymer denominators.

7 First unproved implication and strategic consequence

The target is now a dependent chart–polymer overlap theorem. For the actual regulated link measure $\mu_{t,L}$ and a gauge-invariant exceptional event, one needs activities $z_{t,L}(X)$ satisfying, in the same local-control window,

$$\sup_L \sup_o \sum_{X \ni o} e^{\alpha|X|} |z_{t,L}(X)| < 1. \tag{6}$$

The estimate must retain the following data rather than absorbing them into a generic "large-field error":

- shared-link dependence and nonabelian Bianchi constraints;
- the orbit-space atlas and any Coulomb-slice/Faddeev–Popov density;
- conditional boundary holonomy;
- center flux, torons, instanton-like sectors, and global topology;
- constants uniform in volume, boundary condition, and the repeated RG scale.

The strategic change is sharp. Enlarging the local chart is no longer the only apparent route. The ultraviolet scale already separates the typical curvature, of order $\sqrt{t}$, from a chart radius that remains small enough for nonlinear coercivity. The next work should exploit the actual gauge-covariant dependence structure to prove (6), or else show precisely why the chosen block event and norm fail under conditioning.

8 Frontier audit

Claim	Status	Scope
Poisson-image standalone tail bound	proved	one SU(2) heat-kernel law
Link Haar derivative bounds	proved	local exponential chart
Independent rooted weighted-polymer sum	proved	independent block events
Simultaneous strict witness	restricted	coexistence benchmark
Tail for the full 4D plaquette marginal	open	interacting link measure
Dependent activity bound (6)	open	volume-uniform 4D measure
Repeated RG to a finite physical scale	open	cutoff- and scale-uniform
Continuum construction, OS reconstruction, and gap	open	Clay endpoint

The strongest defensible level is **C2, restricted theorem**. This document does not construct four-dimensional continuum Yang–Mills theory and does not prove a Hamiltonian mass gap. The existence and mass-gap problem remains open.

References

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